

# **PROJECT REPORT**

## **Encryption & Decryption Tool Using Python**

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Year: 2026

### **1. Introduction**

This project focuses on developing a Python-based Encryption and Decryption Tool to ensure data security using cryptographic techniques. The tool allows users to securely encrypt and decrypt text using a secret key. It also includes a logging system to track all operations with timestamps for better monitoring and analysis.

### **2. Objective**

The objective of this project is to understand cryptographic concepts, implement secure encryption and decryption techniques, and build a practical security tool for real-world applications.

### **3. Technologies Used**

- Python
- Cryptography Library (Fernet)
- Command Line Interface (CLI)

### **4. Features**

- Secure key generation
- Text encryption and decryption
- Base64 encoding and decoding
- Hex encoding and decoding
- Caesar cipher implementation
- Automatic output saving functionality
- Timestamp-based logging system
- View saved logs functionality

### **5. Methodology**

The system uses a menu-driven command-line interface where users select operations based on their requirements. For encryption, the input text is processed using a secret key generated through the Fernet algorithm. For decryption, the same key is used to retrieve the original data. Additional encoding techniques such as Base64 and Hex are implemented to enhance functionality and learning. All

operations performed by the user are logged with timestamps, allowing tracking of encryption and decryption activities.

## 6. Output

```
=====
ENCRIPTION & DECRYPTION TOOL
=====
1. Generate Key      2. Fernet Encrypt
3. Fernet Decrypt    4. Base64 Encode
5. Base64 Decode     6. Hex Encode
7. Hex Decode        8. Caesar Encrypt
9. Caesar Decrypt    10. View Saved Logs
11. Exit
=====
Enter choice (1-11): 2
Enter text to encrypt: SecureMessage_Samyak_2026

-----
FERNET ENCRYPT RESULT
-----
Input  : SecureMessage_Samyak_2026
Output : gAAAAABp4maxTqbVXvT7sgdoywsfoNrUkn4ZE2pr0V_hfPeC2oDoK6Ro5mPjC

[SUCCESS] Encrypted output auto-saved in encrypted_output.txt

=====
ENCRIPTION & DECRYPTION TOOL
=====
1. Generate Key      2. Fernet Encrypt
3. Fernet Decrypt    4. Base64 Encode
5. Base64 Decode     6. Hex Encode
7. Hex Decode        8. Caesar Encrypt
9. Caesar Decrypt    10. View Saved Logs
11. Exit
=====
Enter choice (1-11): 3
Enter encrypted text: gAAAAABp4maxTqbVXvT7sgdoywsfoNrUkn4ZE2pr0V_hfPeC2oDoK6Ro5mPjC

-----
FERNET DECRYPT RESULT
-----
Input  : gAAAAABp4maxTqbVXvT7sgdoywsfoNrUkn4ZE2pr0V_hfPeC2oDoK6Ro5mPjC
Output : SecureMessage_Samyak_2026

[SUCCESS] Decrypted output auto-saved in decrypted_output.txt
```

Figure 1: Encryption and Decryption Output

```
=====
ENCRIPTION & DECRYPTION TOOL
=====
1. Generate Key      2. Fernet Encrypt
3. Fernet Decrypt    4. Base64 Encode
5. Base64 Decode     6. Hex Encode
7. Hex Decode        8. Caesar Encrypt
9. Caesar Decrypt    10. View Saved Logs
11. Exit
=====
Enter choice (1-11): 4
Enter text for Base64 encode: PrivateData_Samyak_03

-----
BASE64 ENCODE RESULT
-----
Input  : PrivateData_Samyak_03
Output : UHJpdmF0ZURhdGFfU2FteWFrXzAz

[SUCCESS] Output auto-saved in base64_encoded.txt

=====
ENCRIPTION & DECRYPTION TOOL
=====
1. Generate Key      2. Fernet Encrypt
3. Fernet Decrypt    4. Base64 Encode
5. Base64 Decode     6. Hex Encode
7. Hex Decode        8. Caesar Encrypt
9. Caesar Decrypt    10. View Saved Logs
11. Exit
=====
Enter choice (1-11): 5
Enter Base64 text to decode: UHJpdmF0ZURhdGFfU2FteWFrXzAz

-----
BASE64 DECODE RESULT
-----
Input  : UHJpdmF0ZURhdGFfU2FteWFrXzAz
Output : PrivateData_Samyak_03

[SUCCESS] Output auto-saved in base64_decoded.txt
```

Figure 2: Base64 Encoding and Decoding Output

```
=====
ENCRIPTION & DECRYPTION TOOL
=====
1. Generate Key      2. Fernet Encrypt
3. Fernet Decrypt    4. Base64 Encode
5. Base64 Decode     6. Hex Encode
7. Hex Decode        8. Caesar Encrypt
9. Caesar Decrypt    10. View Saved Logs
11. Exit
=====
Enter choice (1-11): 10
=====
SAVED LOGS
=====

Timestamp : 2026-04-17 22:28:25
Action    : Fernet Encrypt
Status    : SUCCESS
Input     : SecureMessage_Samyak_2026
Output    : gAAAAABp4maxTqbVXvT7sgdoywsfoNzUkn4ZE2pr0V_hfPeC2oDoK6Ro5mPj0gr

Timestamp : 2026-04-17 22:28:44
Action    : Fernet Decrypt
Status    : SUCCESS
Input     : gAAAAABp4maxTqbVXvT7sgdoywsfoNzUkn4ZE2pr0V_hfPeC2oDoK6Ro5mPj0gr
Output    : SecureMessage_Samyak_2026

Timestamp : 2026-04-17 22:29:58
Action    : Base64 Encode
Status    : SUCCESS
Input     : PrivateData_Samyak_03
Output    : UHJpdmF0ZURhdGFuZ2FteWFrXzAz

Timestamp : 2026-04-17 22:30:06
Action    : Base64 Decode
Status    : SUCCESS
Input     : UHJpdmF0ZURhdGFuZ2FteWFrXzAz
Output    : PrivateData_Samyak_03
=====
```

Figure 3: Logs with Timestamp Output

## 7. Applications

- Secure storage of confidential data
- Learning cryptography and data security
- Protecting sensitive communication
- Understanding encryption techniques
- Tracking activities using logs

## 8. Conclusion

This project successfully demonstrates the practical implementation of encryption and decryption using Python. It highlights the importance of data security in modern applications. The addition of logging functionality provides better tracking and monitoring of operations. The project also provides a strong foundation for understanding cryptographic techniques and can be further enhanced with advanced security features.